COURSE OUTLINE



MGMT2501

Introduction to Statistics

Fall 2020 - Fall 2020

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MGMT2501 Introduction to Statistics

COURSE DESCRIPTION

This course explores introductory statistical data analysis and interpretation techniques used in business environments. Learners develop foundational knowledge on key topics including data collection and presentation and measures of descriptive statistics. Random variables, probability and probability distributions, point and interval estimation, hypothesis testing, correlation, and regression analysis are also covered.

REQUISITES	None
EQUIVALENTS	Complete at least 1 of the following courses: • MATH250 - Statistics (4)
CREDITS	3
HOURS	45
ELIGIBLE FOR PLAR	Yes

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Summarize data graphically and numerically	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)
2	Calculate the probabilities of simple and compound events	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)
3	Compute the probabilities of discrete and continuous random variables utilizing probability distributions	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)
4	Compute expected value, variance and co-variance of linear combinations of random variables	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)
5	Apply the central limit theorem to sample mean distributions	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)

6	Employ linear regression and correlation analysis to predict the association between variables	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)
7	Construct and interpret the confidence interval for a population	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)
8	Execute statistical hypothesis tests for a population	Thinking Skills (2), Numeracy and Financial Literacy (3), Digital Literacy (5)

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Module 1: What is Statistics Module 2: Part I: Descriptive Statistic - Graphical Summaries
Week 2	Module 2: Part II: Describing Data: Numerical Measures
Week 3	Module 3: Probability
Week 4	Module 4: Discrete Probability Distributions
Week 5	Module 4: Discrete Probability Distributions
Week 6	Mid-Term Exam
Week 7	Module 5: Part I - Continuous Probability Distributions
Week 8	Module 5: Part II - Sampling Methods and the Central Limit Theorem
Week 9	Reading Week
Week 10	Module 6: Estimation and Confidence Intervals
Week 11	Module 7: One Sample Tests of Hypothesis
Week 12	Module 8: Two-Sample Tests of Hypothesis
Week 13	Module 9: Linear Regression and Correlation
Week 14	Final Exam review and practice
Week 15	Final Exam Week

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1-8	Mid -Exams/Case Study	35%
1-8	In-class group activities	15%
1-8	Online course homework	10%
1-8	Comprehensive Final Exam	40%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

GRADING SCHEME

Grade	Percentage	Grade Point	Description
A+	95-100	4.0	Exceptional: superior knowledge of subject matter
A	90-94	4.0	Excellent: outstanding knowledge of subject matter
A-	85-89	3.67	
B+	80-84	3.33	
B	75-79	3.0	Very Good: knowledge of subject matter generally mastered
B-	70-74	2.67	
C+	67-69	2.33	
C	64-66	2.0	Satisfactory/Acceptable: knowledge of subject matter adequately mastered
C-	60-63	1.67	
D+	57-59	1.33	
D	50-56	1.0	Minimal Pass
F	Less than 50	0.0	Fail: an unsatisfactory performance

REQUIRED LEARNING RESOURCES

Lind D.A., Marchal W.G, Wathen S.T., Waite C.A., (2018). *Basic statistics for business & economics*. (6th ed). Ontario, Canada: McGraw-Hill Ryerson. (PLUS Connect with SmartBook Online Access)

ISBN: 9781259265945 hardcopy plus Connect

ISBN: 9781259269967 eText plus Connect

Open source textbooks (You'll be using this in Module 4) : "OpenIntro Statistics"
https://www.openintro.org/stat/textbook.php?stat_book=os

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.